Monte Carlo Methods and Nuclear Modeling

Introduction to Monte Carlo Modeling
Problem 1

Under the guidance of Professor:
Christoph HARTNACK

ABSTRACT:
The resolution to this problem comprises the development of a **MatLab-GNU Octave** script implementing a Monte-Carlo algorithm, following random point generation and spatial distribution as per "**Von Neumann"** principles [8], to simulate the tri-dimensional collision of nuclei interacting in a projectile-target set-up, to estimate the number of nucleons that participate in the collision.

This modeling proposition was found to be analogous to the "**Nuclear Fireball Model**"[11], and advances a framework to estimate the likelihood of interaction (cross-section) between both nuclei.

In peer-substantiating the aforementioned realizations, it was found that, by assuming "**Werner-Wheeler**" [1] considerations, it is possible to make use of the "**Langevin equation method**" [1] to extend the "**liquid-drop model**"[5] and thus better approximate the behavior of projectile-target interactions as detailed in sections 5.1 and 5.2.

Date	Peer 1	Peer 2	Peer 3
02 Feb 2026	Javier Alonzo ROMO LEON		

The present document formatting has been aligned to meet standard practices as per ISO9001 in support of future standardization processes.

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1. Introduction.

The present document compiles class exercises as taught by lead “Monte Carlo Methods and Nuclear Modeling” teacher, Professor Christoph HARTNACK, with the intent of:

- Documenting resolution methodology,
- Annotate professor-student comments,
- Document lessons learned,
- Document relevant bibliography,
- Provide a quick entry point to solve related scenarios in future professional endeavors.

1.1. Problem summary and location of solutions.

The problem prompts to find a minimum viable algorithm, preferably MatLab-Octave based, that would allow to implement Monte-Carlo analyses to estimate a participant-spectator analysis of two nuclei (e.g. https://doi.org/10.1103/PhysRevLett.37.1202) to assess fundamental nuclear phenomena.

1.2. Source code location and communications.

The MatLab - Octave GNU script developed to resolve this Problem 1 is publicly available at:
https://raw.githubusercontent.com/J4v1987/MatLab-Octave_Nuclear_Modelling/refs/heads/main/Problem1.m

The Git / GitHub full project top-level repository is found in:
https://github.com/J4v1987/MatLab-Octave_Nuclear_Modelling

This MatLab – GNU Octave script has been developed in Ubuntu Linux and requires terminal command.

Should the formatting of plots and figures in this report exhibit suboptimal quality, do consider consulting flat 2D *.png, as well as interactive 3D *.ofig, from the same source code repository:
https://github.com/J4v1987/MatLab-Octave_Nuclear_Modelling/tree/945f312d651e37109ebec891d98a36f7933a736f/Problem1_Plots

Professionals in the nuclear engineering realm and interested audience are invited to participate in the development of this script. Do consider contacting the repository maintainer via dedicated communications found in:
https://sites.google.com/view/b-eng-jarl/home

2. Problem 1 enunciation

Introduction to Nuclear Modelling, Master SNEAM

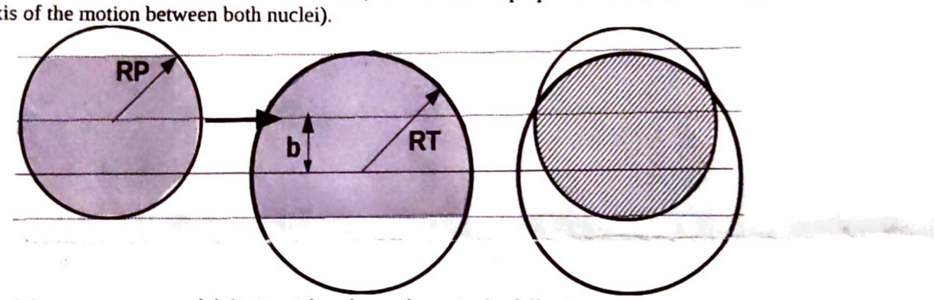
Student laboratories Christoph Hartnack

Problem 1 : Application of the Participant-Spectator-Model **(evaluated)**

In nuclear physics nuclei can be in first order regarded as spheres filled with nucleons. According to the liquid drop model the density inside the nucleus should be independant on the mass number ***A*** and thus the radius ***R*** directly related to ***A***:

$R = R_0\ A^{1/3} \quad R_0 = 1.2$ fm

Imagine the collision of 2 nuclei with the mass numbers ***AP*** and ***AT***. The impact parameter ***b*** describes the transverse distance of the 2 centers, i.e the distance perpendicular to the beam axis (the axis of the motion between both nuclei).



The participant-spectator model distinguishes the nucleons in the following way:

- all nucleons of the projectile/target which are also in the projection (perpendicular to the beam axis) of the other nucleus (target/projectile) are assumed to collide with nucleons of the other nucleus. They participate in the reaction and are thus called **participants**.
- all nucleons that do not fullfill this requirement may not take part in the reaction and are called **spectators**.
- The **participant fraction** is the ratio of the number of participants to the number of all nucleons. In our model we can take this as the ratio of the volume of the participant matter to the volume of both nuclei.

Calculate the participant fraction as a function of impact parameter for the following systems:

1. C+Au, ***AP***=12, ***AT***=197
2. Au+Au ***AP***=***AT***=197

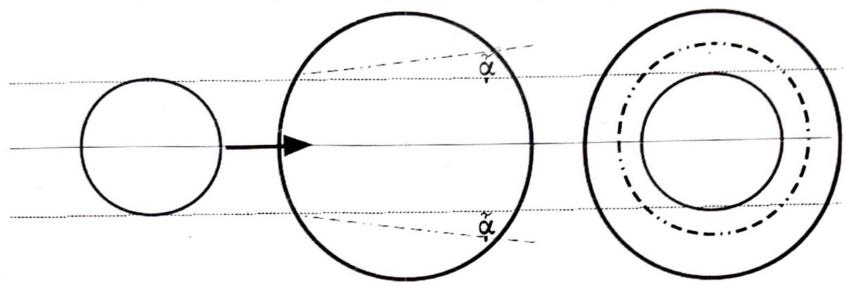
Your work should include the following steps:

1. Calculate the radii of the nuclei and determine the maximum impact parameter
2. Determine seperatly the conditions
   1. For a nucleon of the projetile to be a participant
   2. For a nucleon of the target to be a participant
3. Write a code that calculates the volumes of projectile, target, the projectile participants and the target participants. Calculate the participant fraction as function of b.
4. Plot the participant fraction as function of b for both systems C+Au and Au+Au
5. Discuss your results. Are there special cases with analytic solutions which may help to test whether your code works properly?

Figure 1: Problem enunciation, page 1.

BONUS:
Extension of the participant-spectator model.

In the extension of this simple geometrical model one may imagine that the violent reactions of the participants create a sidewards pressure which may extend into the spectator region.
In this case some kind of cone may be created, characterised by the angle α.



For simplicity we will only regard central ***b***=0 collisions of the system C+Au.
Calculate the participant fraction as a function of the angle α.
Which is the maximum "reasonable" value for α?
Discuss your results. Are there special cases that help to test your code?

Figure 2: Problem enunciation, page 2.

2.1. Problem 1 Interpretation & Script Documentation

[Statement 1] A MatLab - GNU Octave script is to be produced. E.g. https://raw.githubusercontent.com/J4v1987/MatLab-Octave_Nuclear_Modelling/refs/heads/main/Problem1.m

[Statement 2] There are two (2) nuclei, referred to as the target nucleus and the projectile nucleus. As their names imply, the projectile advances at sub-relativistic speed towards the target which is assumed to be static. As seen in problem enunciation Figure 1, the projectile is trajectory is at a distance referred to as the impact parameter, b (same variable in the script).

The script prompts user to instruct an engagement percentage, E_p (in a 0-100 % scale), which is used to scale b such that when E_p=0 then b=0, meaning a direct center-to-center collision between the projectile and target, whereas E_p=100 would mean b equates the arithmetic sum of the projectile and target radii, and thus represent the case where the projectile just 'grazed' the target.

[Statement 3] The script prompts the user to enter both the projectile and target radii (respectively: Ap, At) which the script uses to calculate their nuclei radii (projectile and target: Rp, Rt), assuming $R_0 \approx 1.2$ [fm] [3][5]:

$$R=R_0 \cdot A^{1/3} \text{ [5]}$$

[Statement 4] The script assumes the target is a sphere [10] in euclidean standard space as seen in Figure 3, E^n [7], with center, $C^1(x_1, y_1, z_1)$, coincident with Z^+ (z-positive axis) and sphere surface tangent with the XY plane, explicitly: $x_1=0$, $y_1=0$, $z_1=Rt$. The relation $z_1=Rt$ is referred to in script as *z_off_t*.

The script assumes the projectile is too an sphere [10] in euclidean standard space, E^n [7], with center, $C^2(x_2, y_2, z_2)$, coincident with Z^+ (z-positive axis) and C^2 offset from XY plane by the arithmetic sum of *z_off_t* and b, such that $x_2=[-\infty,\infty]$, $y_2=0$, $z_2=$*z_off_t*+b.

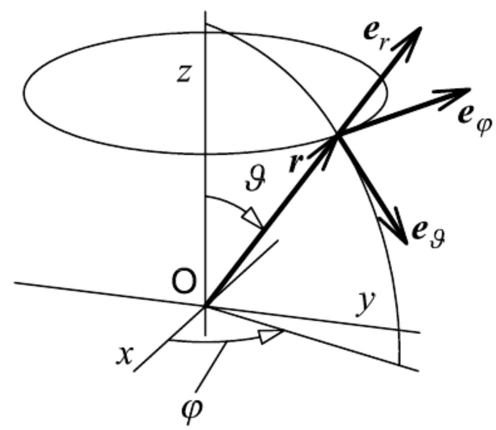


***Figure 3: Right-handed spherical coordinate system in E^n.* [7]**

[Statement 5] Even though the projectile is theoretically in motion, the script uses the instant in time when both projectile C^2 and target C^1 share Z^+ in time and space. After this instance, it is assumed that the projectile will successfully travel through-all the target along E^n x-axis while z_2=*z_off_t*+b persists. If E_p (user prompted) scaling b is <100, then an overlap between projectile and target will occur, which will lead to their corresponding overlapping nucleons to become “participants” while the reminder will be “spectators”.

[Statement 6] The script goes on to fill E^n with a number, *N*, of random points prompted by the script user, each of these summoned using MatLab-Octave built-in function rand() [13], and then mathematically scaled to fit the spatial probability distribution of being generated inside a prismatic volume (script: “for i=1:N”) tightly enclosing both target and projectile. These points are listed in matrix “*prismPointsList*”.

[Statement 7] The script then leverages mathematical principles, like the “Cartesian equation of a sphere and circle” [10][9] to create accept-reject “Von Neumann” [8] volumes whereby prismPointsList points that…

...are within the volume of the target nucleus and within the cylindrical projection along E^n x-axis of the projectile, are deemed volume-wise proportional to target participants (nucleons), and listed in script matrix *targetParticipants*. E. g. Figure 7.

...are within the volume of the projectile nucleus and within the cylindrical projection along E^n x-axis of the target, are deemed volume-wise proportional to projectile participants (nucleons), and listed in script matrix *projectileParticipants*. E. g. Figure 5.

...are within the volume of the target nucleus, are deemed volume-wise proportional to target nucleons, and listed in script matrix *targetNucleons*. E. g. Figure 6.

...are within the volume of the projectile nucleus, are deemed volume-wise proportional to projectile nucleons, and listed in script matrix *projectileNucleons*. E. g. Figure 4.

[Statement 8] The script uses all aforementioned, along with key MatLab-Octave built-in functions like *length()* [12] to obtain the participants fraction, N_p, which details the total number of participants, known as the “**Nuclear Fireball Model**” [11], in comparison to the total number of nucleons. This along other calculations are issued at end of script runtime as exemplified in 3.2 Octave terminal outputs (Case: C+Au). User inputs are collected as exemplified in 3.1 Octave usage (Case: C+Au).

[Statement 9] Finally, the script produces 3D plots in the same location where the script is stored and executed, namely:
Problem1 - target participants,
Problem1 - projectile mc points,
Problem1 - projectile participants
Problem1 - target mc points,

Each of the aforementioned is saved in a static 3D plot (*.png file) and an interactive 3D plot (*.ofig).

3. Solutions to Problem 1

3.1. Octave usage (Case: C+Au)

>> clear all
>> close all
>> clc
>> Problem1
Projectile atomic mass [u] or [g·mol^{-1}]: 12
Target atomic mass [u] or [g·mol^{-1}]: 197
Projectile-target impact parameter scaling factor (100: just disengaged, 0: center to center): 80
Number Monte Carlo trials (e.g. 500): 10000
Do you want to see the analysis 3D plot? (may take longer to process)(yes or no) yes

PROCESSING…

3.2. Octave terminal outputs (Case: C+Au)

3.2.1.[00. ATOMIC MASSES]

Ap: Projectile particle atomic mass: 12 [u]
At: Target particle atomic mass: 197 [u]

3.2.2.[01. NUCLEI RADII & MAXIMUM IMPACT PARAMETER]

∵ $R=R_0 \cdot A$^1/3 (Kenneth S Krane. ISBN: 0-471-80553-X, page 48.)
∧ R_0= 1.20 [fm] (Kenneth S. Krane. ISBN: 0-471-80553-X, page 48.)
∧ A= 12.00 [u] (user input, script variable Ap)
⟹ R(Ap)=Rp= 2.75 [fm]

∵ $R=R_0 \cdot A$^1/3 (Kenneth S. Krane. ISBN: 0-471-80553-X, page 48.)
∧ R_0= 1.20 [fm] (Kenneth S. Krane. ISBN: 0-471-80553-X, page 48.)
∧ A= 197.00 [u] (user input, script variable Ap)
⟹ R(At)=Rt= 6.98 [fm]

∵ "...the impact parameter (the perpendicular distance from the center of the target nucleon to the line of flight of the incident nucleon). See: Kenneth S. Krane. ISBN: 0-471-80553-X, page 86.
→b=Rp+Rt (maximum impact parameter, b, equals the projectile radius, Rp, plus the target radius, Rt.
⟹ b=9.73 [fm] max.

Note: based in user prompt, b has been scaled to 80.00 %, where 0% denotes center-to-center impact, whereas 100% denotes maximum impact parameter b (projectile particle just grazed target particle).

The effective impact parameter in the analysis is: b = 7.78 [fm]

3.2.3.[02. PARTICIPANT-SPECTATOR FRAMEWORK]

While considering the "Nuclear Fireball Model" (see: https://doi.org/10.1103/PhysRevLett.37.1202), where the projectile particle collides through-all in straight trajectory line with the target particle, it then follows that "...The swept-out nucleons from the projectile and target are called the fireball...", where the fireball would constitute the participant nucleons, while the spectators would be the remainder nucleons.

Let there be a right-handed Cartesian coordinate system in standard Euclidean space as per ISO 80000-2:2009(E), E^n:

- The target particle of radius Rt= 6.98, is envisioned still, with center at (0,0,6.98)
- The projectile particle of radius Rp= 2.75, is envisioned as displacing at sub-relativistic speed along E^n x-axis. For Monte-Carlo analysis purposes it is envisioned to be in an instant of time with its center at (0,0,14.77).
- This array allowed to randomly scatter N=10000.00 points within a prismatic volume tightly enveloping both the projectile and the target particles.

3.2.4.[03. VOLUMETRIC ANALYSIS]

ANALYTIC VOLUMES:
Monte Carlo test volume: 489.14 [fm^3]
Projectile volume: 86.86 [fm^3]
Target volume: 1425.93 [fm^3]

MONTE CARLO POINTS:
Monte Carlo test points (encompassing the volumes of projectile and target), N: 10000.00 [points]
Projectile monte carlo points (assumed to be proportional to all projectile nucleons): 266.00 [points]
Target monte carlo points (assumed to be proportional to all target nucleons): 4205.00 [points]

PARTICIPANT POINTS:
Projectile participant monte carlo points: 82.00 [points]
Target participant monte carlo points: 119.00 [points]

PARTICIPANT FRACTION (Impact parameter, b=7.78 [fm] implicit):

Let:
Participant fraction: N_p
Target participant Monte Carlo points: TPP=119.00 [points]
Projectile participant Monte Carlo points: PPP=82.00 [points]
Projectile total Monte Carlo points: PMP=266.00 [points]
Target total Monte Carlo points: TMP=4205.00 [points]

It follows that:
∵ "...The participant fraction is the ratio of the number of participants to the number of all nucleons..." (C. Hartnack, "Student Laboratories, Master SNEAM, Application of the Participant-Spectator-Model, Problem 1")
→ N_p = (TPP+PPP)/(PMP+TMP)

$\Longrightarrow N_p = 0.04$

3.2.5. [04. Plots (C+Au)]

Projectile: Carbon (C) in blue-cyan.
Target: Gold (Au) in red-magenta.

3.2.5.1. (C) Projectile Points

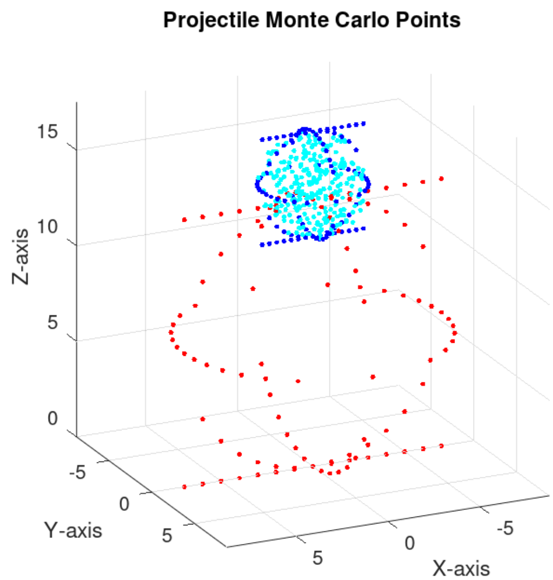


Figure 4: Carbon projectile Monte Carlo points

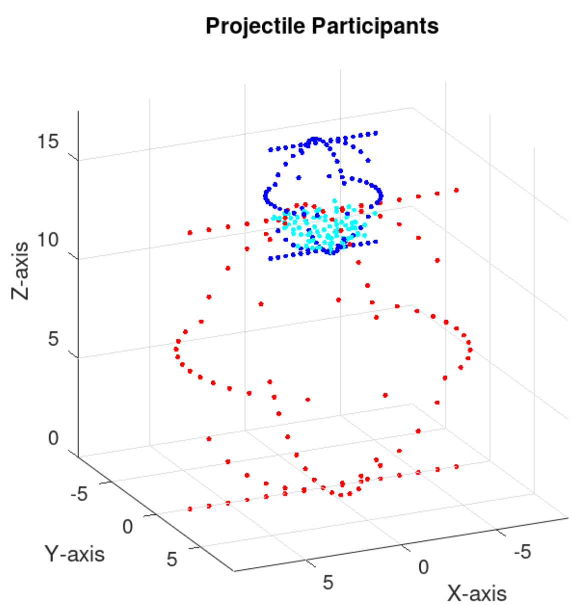


Figure 5: Carbon projectile participant points

3.2.5.2. (Au) Target Points

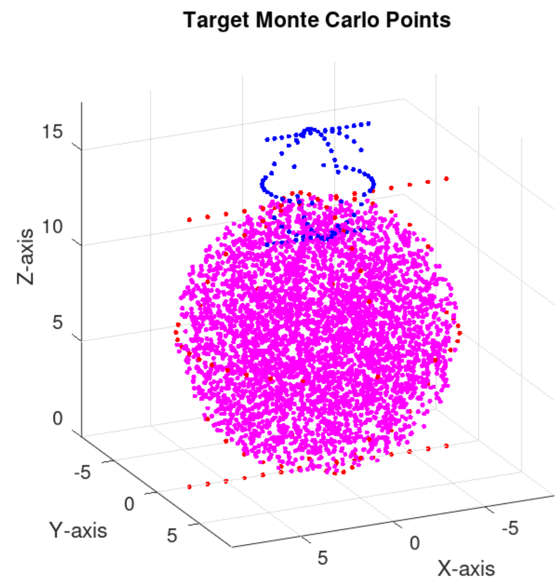


Figure 6: Carbon target Monte Carlo points

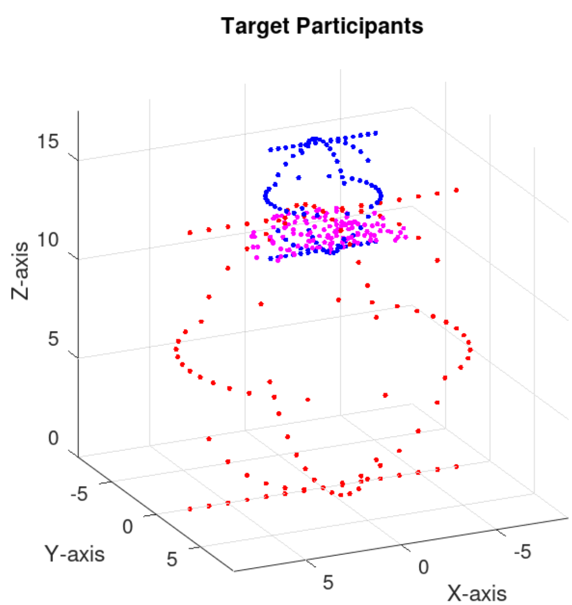


Figure 7: Carbon target participant points

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3.3. Octave usage (Case: Au+Au)

>> clear all
>> close all
>> clc
>> Problem1
Projectile atomic mass [u] or [g·mol^{-1}]: 197
Target atomic mass [u] or [g·mol^{-1}]: 197
Projectile-target impact parameter scaling factor (100: just disengaged, 0: center to center): 80
Number Monte Carlo trials (e.g. 500): 10000
Do you want to see the analysis 3D plot? (may take longer to process)(yes or no) yes

PROCESSING…

3.4. Octave terminal outputs (Case: Au+Au)

3.4.1.[00. ATOMIC MASSES]

Ap: Projectile particle atomic mass: 197 [u]
At: Target particle atomic mass: 197 [u]

3.4.2.[01. NUCLEI RADII & MAXIMUM IMPACT PARAMETER]

∵ $R=R_0 \cdot A^{1/3}$ (Kenneth S Krane. ISBN: 0-471-80553-X, page 48.)
∧ R_0= 1.20 [fm] (Kenneth S. Krane. ISBN: 0-471-80553-X, page 48.)
∧ A= 197.00 [u] (user input, script variable Ap)
⟹ R(Ap)=Rp= 6.98 [fm]

∵ $R=R_0 \cdot A^{1/3}$ (Kenneth S. Krane. ISBN: 0-471-80553-X, page 48.)
∧ R_0= 1.20 [fm] (Kenneth S. Krane. ISBN: 0-471-80553-X, page 48.)
∧ A= 197.00 [u] (user input, script variable Ap)
⟹ R(At)=Rt= 6.98 [fm]

∵ "...the impact parameter (the perpendicular distance from the center of the target nucleon to the line of flight of the incident nucleon). See: Kenneth S. Krane. ISBN: 0-471-80553-X, page 86.

→b=Rp+Rt (maximum impact parameter, b, equals the projectile radius, Rp, plus the target radius, Rt.
⟹ b=13.96 [fm] max.

Note: based in user prompt, b has been scaled to 80.00 %, where 0% denotes center-to-center impact, whereas 100% denotes maximum impact parameter b (projectile particle just grazed target particle).

The effective impact parameter in the analysis is: b = 11.17 [fm]

3.4.3.[02. PARTICIPANT-SPECTATOR FRAMEWORK]

While considering the "Nuclear Fireball Model" (see: https://doi.org/10.1103/PhysRevLett.37.1202), where the projectile particle collides through-all in straight trajectory line with the target particle, it then follows that "...The swept-out nucleons from the projectile and target are called the fireball...", where the fireball would constitute the participant nucleons, while the spectators would be the remainder nucleons.

Let there be a right-handed Cartesian coordinate system in standard Euclidean space as per ISO 80000-2:2009(E), E^n:

- The target particle of radius Rt= 6.98, is envisioned still, with center at (0,0,6.98)
- The projectile particle of radius Rp= 6.98, is envisioned as displacing at sub-relativistic speed along E^n x-axis. For Monte-Carlo analysis purposes it is envisioned to be in an instant of time with its center at (0,0,18.15).
- This array allowed to randomly scatter N=10000.00 points within a prismatic volume tightly enveloping both the projectile and the target particles.

3.4.4.[03. VOLUMETRIC ANALYSIS]

ANALYTIC VOLUMES:
Monte Carlo test volume: 702.05 [fm³]
Projectile volume: 1425.93 [fm³]
Target volume: 1425.93 [fm³]

MONTE CARLO POINTS:
Monte Carlo test points (encompassing the volumes of projectile and target), N: 10000.00 [points]
Projectile monte carlo points (assumed to be proportional to all projectile nucleons): 2864.00 [points]
Target monte carlo points (assumed to be proportional to all target nucleons): 2943.00 [points]

PARTICIPANT POINTS:
Projectile participant monte carlo points: 219.00 [points]
Target participant monte carlo points: 234.00 [points]

PARTICIPANT FRACTION (Impact parameter, b=11.17 [fm] implicit):

Let:
Participant fraction: N_p
Target participant Monte Carlo points: TPP=234.00 [points]
Projectile participant Monte Carlo points: PPP=219.00 [points]
Projectile total Monte Carlo points: PMP=2864.00 [points]
Target total Monte Carlo points: TMP=2943.00 [points]

It follows that:
∵ "...The participant fraction is the ratio of the number of participants to the number of all nucleons..." (C. Hartnack, "Student Laboratories, Master SNEAM, Application of the Participant-Spectator-Model, Problem 1")
→ N_p = (TPP+PPP)/(PMP+TMP)

⟹ N_p = 0.08

3.4.5. [04. Plots (Au+Au)]

Projectile: Gold (Au) in blue-cyan.
Target: Gold (Au) in red-magenta.

3.4.5.1. (Au) Projectile Points

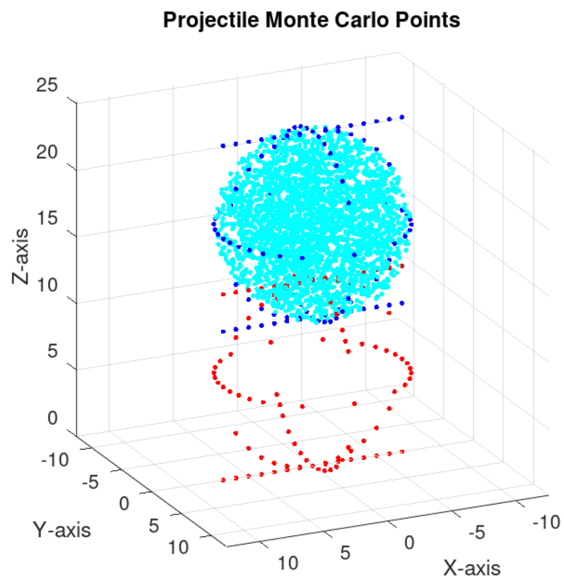

Figure 8: Gold projectile Monte Carlo points

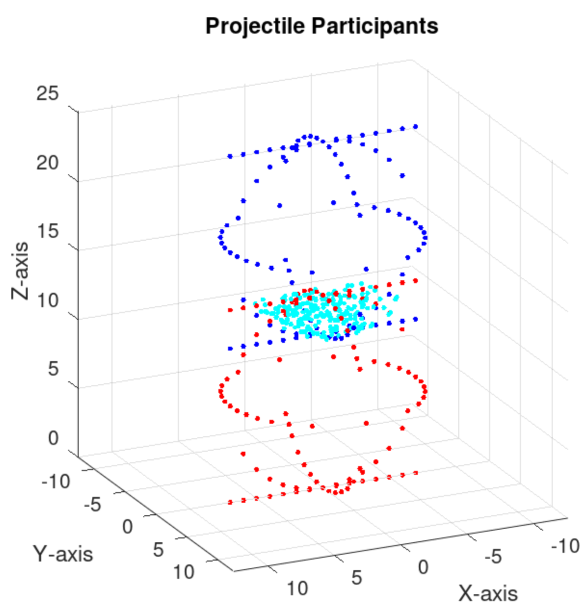

Figure 9: Gold projectile participant points

3.4.5.2. (Au) Target Points

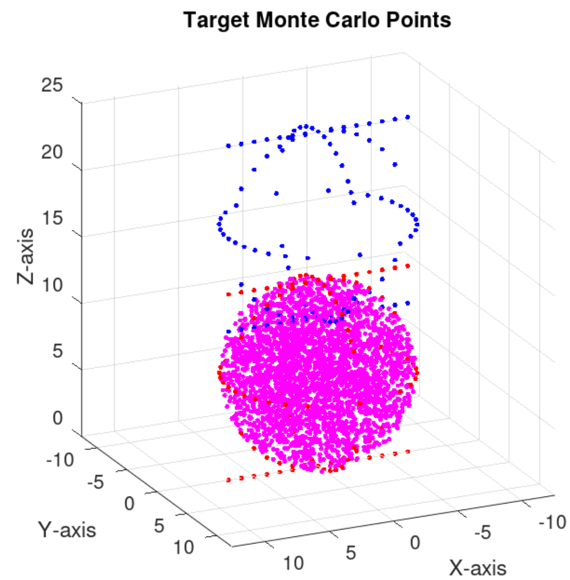

Figure 10: Gold target Monte Carlo points

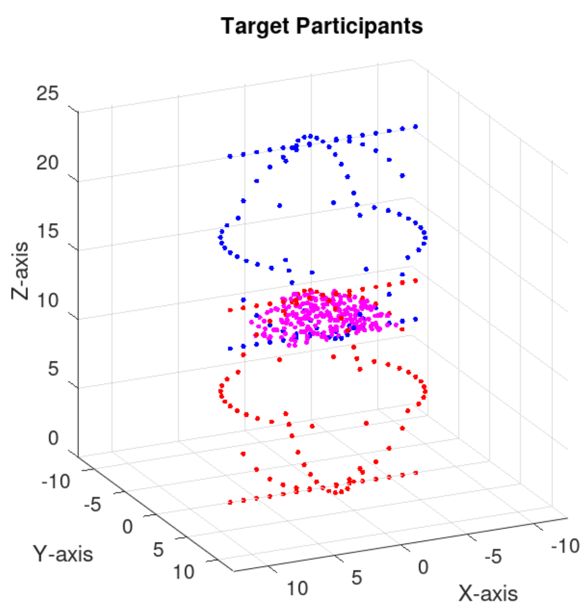

Figure 11: Gold target participant points

4. Discussion & Conclusions

[Conclusion 1] It is recommended to iterate more than thirty (30) times each Monte Carlo experiment to improve the statistical accuracy of nuclei participants, N_p, and thus report it instead as a mean number of participants, $\mu(N_p)$, which would need to be accompanied with their standard deviations.

[Conclusion 2] The simulation of the "Nuclear Fireball Model" down to the level of individual nucleons shall be averted, for these follow complex probabilistic rules detailed by authors like Krane in reference [5], conditioned in space and frequency, that would render their modeling by means of the Von Neumann [8] accept-reject technique impossible.

[Conclusion 3] The parametric coding of the script as detailed in [Statement 5], was found to be a good practice for it enabled the script to accept nuclei of varying mass numbers Ap, At, as well as a range of impact parameters, b via E_p, to seamlessly try different collision set-ups without the need of revising the script MatLab–Octave source code.

[Conclusion 4] From [Conclusion 3] one can derive that the script may be considered an efficient tool to estimate nuclear cross-sections because of E_p.

[Conclusion 5] The script could be further improved by implementing [Conclusion 1] in the form of an additional MatLab-Octave terminal prompt to user that would feed a for-loop to iterate the script up to a statistically-significant number of times depending on the user's hardware computing speed and precision requirements.

5. BONUS

One way to extend the participant-spectator model would be to envision the sidewards pressure deforming the array of nucleons in through-all collisions, inflicting a hole through-all the target with a beveling effect because of the distribution of impact energies.

Such mathematical and programmatic feat would require to take into account nuclei properties thoroughly detailed by authors like Krane in reference [5], such as:

- Nucleon-nucleon binding energies.
- Strong-forces.
- Coulomb repulsion forces.
- The “liquid-drop model”.
- Nucleus shell model.

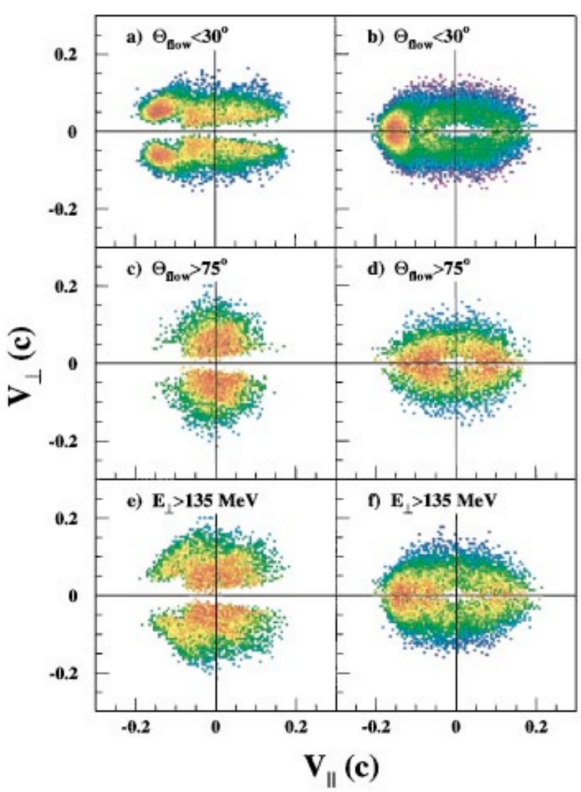

Figure 12: (left) cross-section views of a target being pierced through-all by a high-speed projectile. [2]

From the initial Von Neumann propositions in 1951 [8] to date, there have been scientific experiments which have advanced a model in this regard.

5.1. “High-speed” projectile scenario

From research conducted among world scientists, it was found that modern ‘state of the art’ simulations account for scenarios where the projectile speed is so high, that the interplay of strong-forces and Coulomb repulsion forces are slow enough to permit the forming of a hole through-all the target that exhibits the intuitively anticipated beveling effect as shown in Figure 12 as per reference [2], furthered in reference [4].

5.2. "Low-speed" projectile scenario

Conversely, if the projectile speed is "slow" enough, the interplay of strong-forces and Coulomb repulsion forces will catch-up with the speed of the projectile, causing the target to indeed behave like a liquid-drop, immediately "healing" the projectile-inflicted forming hole as it passes-through the target. Furthermore, as the projectile departs from the target, it is presently believed that a target-projectile 'neck' of nucleons will form.

Parameters like the macroscopic Werner-Wheeler mass parameter enable the use of the unified Langevin equation method to model the geometry of such 'neck' of nucleons [1] as shown in Figure 13, where:

- D_o: an experimental scaling coefficient, "…approximately 1.19 ($\sim 2^{1/4}$)…" [1],
- ρ: "…is the nucleus density…" [1],
- σ: "...surface tension per unit area…" [1],
- t: time [1],
- R_o: "...the nuclear radius with mass number, A, before the collisions…" [1],
- R_{neck}: "...the neck radius…" as a function of time, t [1].

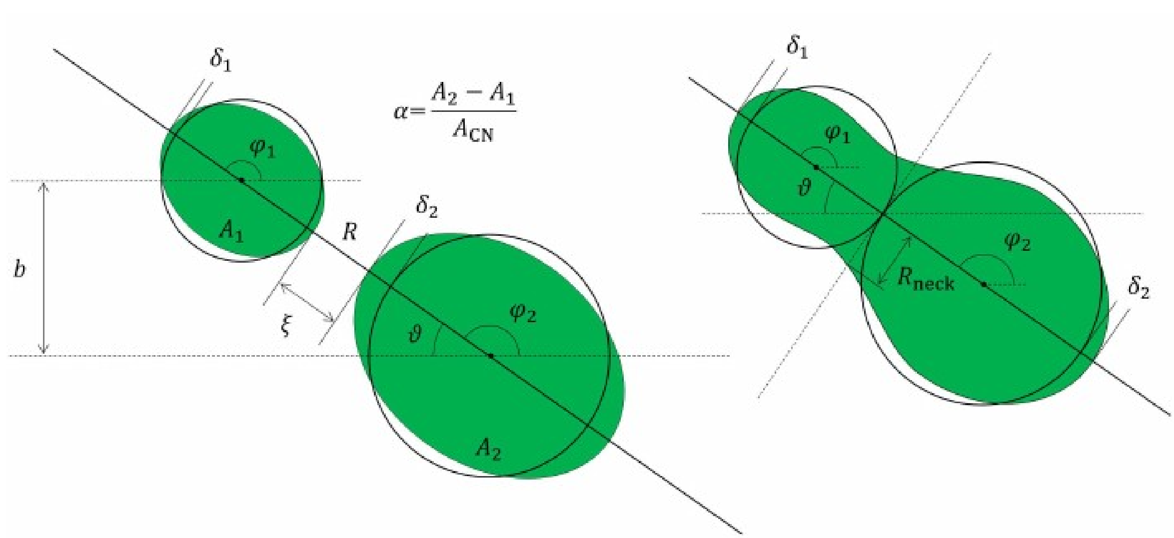

Figure 13: Neck geometry [1]

$$R_{neck}(t) \approx \left(D_o \cdot \left\{ \left[(\sigma \cdot R_o) / \rho \right]^{0.25} \right\} \right) \cdot \left\{ \mathrm{sqrt}\left(t^{56.57} \right) \right\} \; [1]$$

6. Comments & Observations from Peer-reviewer(s)

7. References

[1] Amano, S., Aritomo, Y. and Ohta, M. (2023). Nuclear viscosity estimated by dynamics of neck formation in the early stage of nuclear collision. Modern Physics Letters A, [online] 38(30n31). doi:https://doi.org/10.1142/s0217732323501365.

[2] Beaulieu, L., Larochelle, Y., Gingras, L., Ball, G.C., Bowman, D.R., Djerroud, B., Doré, D., Galindo-Uribarri, A., Guinet, D., Hagberg, E., Horn, D., Laforest, R., Lautesse, P., Roy, R., Samri, M. and St-Pierre, C. (1996). Direct Measurement of Dissipation in the 35Cl+12C reaction at 43 MeV/nucleon. Physical Review Letters, 77(3), pp.462–465. doi:https://doi.org/10.1103/physrevlett.77.462.

[3] Bureau International des Poids et Mesures (2024). The International System of Units (SI). 9th ed. [online] Bureau International des Poids et Mesures. Available at: https://www.bipm.org/documents/20126/41483022/SI-Brochure-9.pdf [Accessed 31 Jan. 2026].

[4] Di Toro, M., Olmi, A. and Roy, R. (2006). Neck dynamics. The European Physical Journal A, [online] 30(1), pp.65–70. doi:https://doi.org/10.1140/epja/i2006-10106-9.

[5] Krane, K.S. and Halliday, D. (1988a). Introductory nuclear physics. New York: John Wiley, p.48. Formula 3.8.

[6] Krane, K.S. and Halliday, D. (1988b). Introductory nuclear physics. New York: John Wiley, p.86.

[7] Technical Committee SS/7, General metrology, quantities, units and symbols. (2010). Quantities and Units. Mathematical Signs and Symbols to Be Used in the Natural Sciences and Technology. [online] British Standards. Available at: https://cdn.standards.iteh.ai/samples/31887/0add63c0863f416e8ce389ed9edd12ad/ISO-80000-2-2009.pdf [Accessed 31 Jan. 2026].

[8] Von Neumann, J. (1951). Various techniques used in connection with random digits. Applied Math Series, [online] 12, pp.36–38. Available at: https://mcnp-green.lanl.gov/pdf_files/InBook_Computing_1961_Neumann_JohnVonNeumannCollectedWorks_VariousTechniquesUsedinConnectionwithRandomDigits.pdf [Accessed 2 Feb. 2026].

[9] Weisstein, E.W. (n.d.). Circle. [online] mathworld.wolfram.com. Available at: https://mathworld.wolfram.com/Circle.html.

[10] Weisstein, E.W. (n.d.). Sphere. [online] mathworld.wolfram.com. Available at: https://mathworld.wolfram.com/Sphere.html [Accessed 26 Jan. 2026].

[11] Westfall, G.D., Gosset, J., Johansen, P.J., Poskanzer, A.M., Meyer, W.G., Gutbrod, H.H., Sandoval, A. and Stock, R. (1976). Nuclear Fireball Model for Proton Inclusive Spectra from Relativistic Heavy-Ion Collisions. Physical review letters, 37(18), pp.1202–1205. doi:https://doi.org/10.1103/physrevlett.37.1202.

[12] The Octave-Forge Community (2017a). Function Reference: length. [online] Sourceforge.io. Available at: https://octave.sourceforge.io/octave/function/length.html [Accessed 1 Feb. 2026].

[13] The Octave-Forge Community (2017b). Function Reference: rand. [online] Sourceforge.io. Available at: https://octave.sourceforge.io/octave/function/rand.html [Accessed 1 Feb. 2026].